



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2019-0186



Agricultural Machinery – Disc Plow GDP 320

Test Report Number	2019-0186
Date Issuance	July 29, 2019
Valid Until	July 29, 2024
Agricultural Machinery	Disc Plow
Type	Tractor-mounted, 3-bottom, one-way disc plow
Brand	GDP
Model	320
Test Requested by	GO TRAKTORA TRADING John Deere Bldg. McArthur Hi-way, Brgy. Matatalaib, Tarlac City, Tarlac

SUMMARY

Construction Requirements	Based on PAES 121:2001	AMTEC Test
Type of disc	Plain or notched	Plain
Number of disc	2 to 8	3
Diameter of discs, mm	560 to 810	515
Working width, mm	500 to 2000	1025.9
Working depth, mm	200 to 400	92.6
Frame height, mm	650 to 800	665
Disc spacing, mm	500 to 750	457
Weight per disc*, kg	150 to 250	ND
Drawbar power requirement per disc, kW	15 to 20	NM
Thickness, mm	Not specified	5
Concavity, mm	Not specified	55

Note: * It is the weight of the plow divided by the number of discs.

OBSERVATIONS

1. Operator's Manual	Only technical specifications brochure was provided.
2. Previous Valid Test Reports of Similar Brand and Model	None
3. Four-wheel tractor used	33.56 kW FARM ROVER 4145 Four wheel tractor
4. Number of operator/s	1
5. Manufacturer	SHANDONG YUNTAI MACHINERY CO., LTD
6. Handling and stability when machine is working	The tractor was stable during plowing.
7. Handling and stability when machine is turning	The tractor was stable during turning.
8. Straightness of furrow	The tractor maintained a straight path during plowing.
9. Quality of soil inversion	Inversion of furrow slice was observed.
10. Uniformity of depth, mm	90 ± 10
11. Non-adhesion of soil to disc	Non-adhesion of soil was observed.
12. Making of adjustments	Adjustments on the depth of tillage can be done.
13. Durability of parts (based on wear of soil-working parts, visible deformation, etc)	No deformation observed during testing.
14. Other Remarks	No breakdown occurred during the test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date/s of Test	April 6, 2019
Location of Test	Brgy San Roque, Tarlac City
Items Tested	Field Capacity, Field Efficiency, Fuel Consumption Rate, Noise Level, Width and Depth of Cut, and Number of Operators

METHODS OF TEST

The machine was tested based on PNS/PAES 132:2004 – Agricultural Machinery – Disc/Moldboard Plow – Methods of Test.

DESCRIPTION OF THE MACHINE

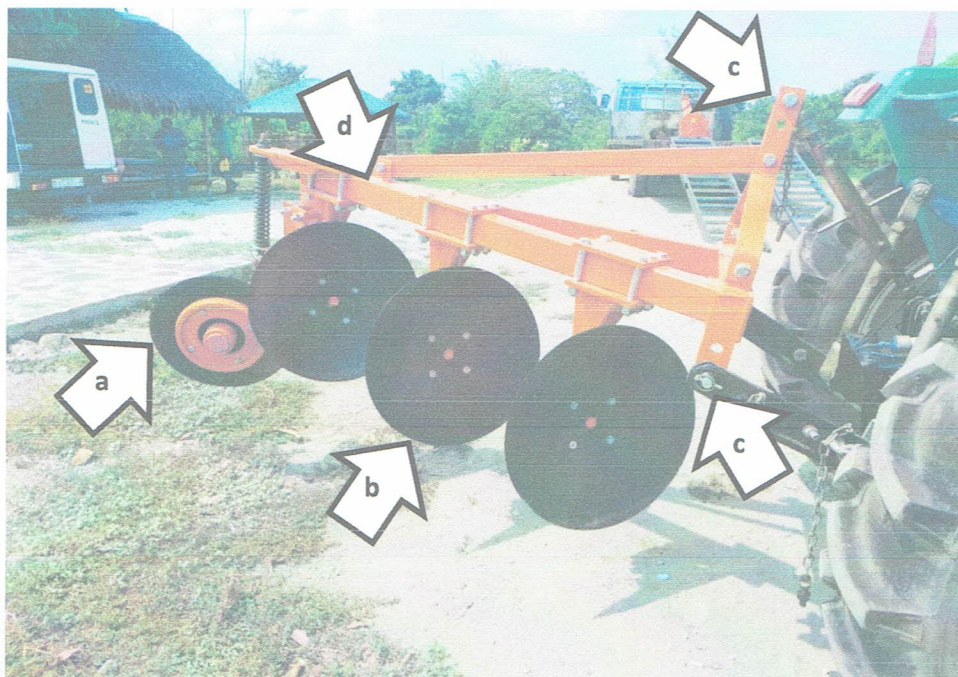


Figure1. Parts of GDP 320 3-Bottom Disc Plow hitched to FARM ROVER 4145 Four-wheel Tractor: a) Rear thrust wheel, b) Disc, c) Hitch point, and d) Frame

SPECIFICATIONS

ITEMS	Manufacturer's Specification	Verification by AMTEC
1 Dimensions and weight		
1.1 Overall length, mm	ND	1722
1.2 Overall width, mm	ND	695
1.3 Overall height, mm	ND	1060
1.4 Weight, kg	ND	NM
1.5 Weight per disc, kg	ND	NM
2 Disc plow		
2.1 Number of discs	ND	3
2.2 Brand	ND	GDP
2.3 Model	ND	320
2.4 Make	ND	China
2.5 Type (plain or notched)	ND	Plain
2.6 Diameter	ND	515
2.7 Thickness, mm	ND	5
2.8 Concavity, mm	ND	55
2.9 Disc spacing, mm	ND	457
2.10 Side angle, °	ND	NM
2.11 Tilt angle, °	ND	NM
2.12 Scraper		
2.12.1 Width, mm	ND	NA
2.12.2 Thickness, mm	ND	NA
2.13 Main frame		
2.13.1 Dimension, mm		
2.13.1.1 Tubular steel	ND	61 × 80
2.13.1.2 C-bar	ND	64 × 100
2.13.2 Frame height, mm	ND	665
2.13.3 Material	ND	C-bar and Tubular steel
2.14 Rear furrow wheel		
2.14.1 Diameter, mm	ND	430
2.14.2 Thickness, mm	ND	4

Note: ND – No Data, NM – Not Measured, NA – Not Applicable

RESULTS

ITEMS	DATA
1 Test Conditions	
1.1 Condition of field	
1.1.1 Location	Brgy. San Roque, Tarlac City
1.1.2 Dimensions of field (L x W), m	46 x 23
1.1.3 Area, m ²	1058
1.1.4 Soil type	ND
1.1.5 Moisture content, %	NM
1.1.6 Weed density	Low
1.1.7 Soil hardness, kg/cm ²	5.51
1.1.8 Last crop planted	Palay
2 Field Performance	
2.1 Date of test	April 6, 2019
2.2 Type of field operation	Plowing, Primary
2.3 Tractor's gear shift setting	3 rd Gear – Low & 4 th Gear – Low
2.4 Traveling or operating speed, kph	4.80
2.5 Depth of tillage, mm	92.6
2.6 Width of tillage, mm	1025.9
2.7 Time lost, min	
2.7.1 Turning	-
2.7.2 Others	0
2.8 Duration of test, min	24.8
2.9 Actual field capacity	
2.9.1 ha/h	0.256
2.9.2 h/ha	3.91
2.10 Theoretical field capacity	
2.10.1 ha/h	0.334
2.10.2 h/ha	2.99
2.11 Field efficiency, %	76.6
2.12 Fuel consumption rate, L/h	5.43
2.13 Effective fuel consumption, L/ha	22.7
2.14 Method of operation	Headland pattern from back furrow
2.15 Percent unplowed/overlap	24.5
2.16 Noise level, dB(A)	86.6

Note: ND – No Data, NM – Not Measured

OBSERVATIONS



Figure 2. Quality of work

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JUL 29 2019
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REMINDER

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